The Gate Knows, the Threshold Doesn't: Label-Free Recalibration of Activation-Gated Jailbreak Defenses Under Unseen Framings

Anonymous submission

Abstract

Activation-gated defenses read a language model's residual stream, decide whether a prompt is harmful, and only then intervene, for example by steering towards refusal. They are attractive because they are cheap and leave benign traffic untouched, but they are calibrated on the jailbreak framings seen during development. We evaluate five such gates on real harmful and benign corpora wrapped in eight framing families, four of which are held out entirely. On unseen framings recall collapses: at a 1% benign false-positive budget, every gate detects at most 17% of harmful fiction or indirect-injection prompts, and the undefended attack success rate of 89% on fiction is left unchanged. The information has not disappeared. A probe that reads the mean of the user's tokens still ranks harmful above benign prompts with AUROC 0.98–0.99 on these framings, and re-thresholding it on the new framing would recover 79–81% recall. The failure is a calibration shift, not a representation failure. We turn this diagnosis into a remedy that needs no labels: cluster the incoming prompt stream by framing, standardise the gate score robustly within each cluster, and fix a single threshold on validation streams. Selected by leave-one-framing-out on validation data and evaluated once on the test split, the method lifts recall on unseen fiction prompts from 7% to 87% and on indirect injection from 17% to 65% at a 5% harmful prevalence, with benign false positives near the budget on these framings. At generation level, it cuts the judged attack success rate of fiction prompts from 89% to 34% and raises benign over-refusal by less than one point. We also report where it fails: when harmful prompts make up a fifth or more of the traffic, when the target model cannot parse the framing, and when a framing cluster is pure noise.

1 Introduction

Safety training teaches a language model to refuse harmful requests, and jailbreaks teach it to forget. A request that is refused when asked plainly is often answered when it is wrapped in a persona, a story, a retrieved document, or an encoding [10, 18, 35, 40]. Because these wrappers change the surface of the prompt but not what is being asked, a natural line of defense looks beneath the surface. Residual-stream activations carry a linearly readable signal of harmfulness [3, 45, 47], and several defenses read that signal to gate an intervention: conditional activation steering adds a refusal direction only when a condition vector fires [16], and probe-based detectors flag a prompt before generation [23, 44]. Such gates are appealing in deployment. They cost one forward pass, they leave benign traffic alone, and their operating point can be set to a fixed false-positive budget.

This paper is about what happens to that operating point when the framing changes. A gate is trained and thresholded on the framings its developers anticipated. Attackers do not oblige. We therefore separate framing families into a *seen* set used for training and calibration and an *unseen* set that is never touched until the final test, and we evaluate on real prompts from HarmBench, AdvBench, JailbreakBench, XSTest and OR-Bench [6, 8, 22, 31, 48], with generations judged by a guard model [30] rather than by refusal string matching.

The first finding is a collapse. Every gate we test, including a geometric gate of our own design that explicitly tries to factor framing out of the representation, keeps more than 80% recall on seen framings and drops to between 0% and 17% on fiction and indirect injection. At the level of generated text, none of them lowers the 89% attack success rate of the fiction framing at all.

The second finding is that the collapse is not where one would expect. If the harmful signal were destroyed by the framing, no threshold could recover it. Instead, the area under the ROC curve barely moves: a probe

reading the user's tokens separates harmful from benign fiction prompts with AUROC 0.99. What moves is the location of both score distributions. A threshold set on seen framings lands in the wrong place, and the same probe re-thresholded on the new framing detects 80% of harmful fiction prompts. Concurrent work reaches compatible conclusions in related settings: linear jailbreak probes are poorly calibrated off-distribution [15], and distribution shift degrades threshold transfer before it degrades ranking [21]. We confirm the pattern systematically across gates, readout positions and framings, and we show that the most principled representational fix we could build, projecting out a subspace of framing directions, does not repair it.

The third contribution is a remedy that targets the calibration directly and requires no labels. Deployed gates see a stream of mostly benign prompts. We cluster that stream by framing features, standardise the gate score inside each cluster with median and median absolute deviation, which are insensitive to a small fraction of harmful prompts [17, 32], and apply one threshold that is fixed on validation streams. The idea of recalibrating subpopulations without labels has antecedents in general machine learning [12, 46]; what we contribute is its adaptation to activation-gated safety, the choice of features to cluster on, a selection protocol that never looks at the unseen framings, and an honest account of the regimes in which it breaks.

In summary, we make four contributions:

- A held-out-framing benchmark for activation-gated defenses on real corpora, with leave-one-framing-out model selection and a single evaluation on the test split.
- Evidence that gate failure under unseen framings is a calibration shift, including a negative result for framing-subspace projection.
- A label-free, framing-adaptive recalibration that restores most of the recall an oracle threshold would give, at detection and at generation level.
- A characterisation of its limits: harmful prevalence, framings the model cannot parse, and adaptive attackers.

2 Related Work

Jailbreak framings. Jailbreaks range from optimised adversarial suffixes [20, 48] and iterative attacker models [7, 24] to hand-written templates such as role-play personas [35], nested fiction [18], encodings and ciphers [40, 42], many-shot demonstrations [2], and instructions planted in retrieved content [10]. Wei et al. [40] attribute their success to competing objectives and mismatched generalisation; the second mechanism is exactly what a framing held out from training probes.

Representation-level defenses. Refusal in chat models is mediated by a low-dimensional direction [3], and harmfulness and refusal are encoded separately and at different token positions [45]. Representation engineering reads and edits such directions [47]; circuit breakers reroute harmful representations during training [49]; latent adversarial training hardens them [36]. Closest to our setting are inference-time gates: CAST steers conditionally on a learned condition vector [16], and probe- or gradient-based detectors flag prompts before decoding [13, 25, 41, 44]. Several recent studies question how far these signals generalise. Probes for high-stakes interactions keep high AUROC out of distribution while recall at a low false-positive rate is modest [23]; final-token probes miss jailbreaks whose evidence sits in earlier user tokens [9]; linear jailbreak probes conflate benign register with intent and their headline accuracy depends on the threshold [15]; and internal harmfulness scores can even anti-rank successful jailbreaks [21]. Adaptive attacks that optimise against the probe itself can drive recall to zero [5]. We study a complementary, non-adaptive failure, natural framing shift, and propose a remedy for it.

Guard models and judges. Classifier guards such as Llama Guard, WildGuard, ShieldGemma, Granite Guardian and Qwen3Guard moderate inputs and outputs [11, 14, 26, 30, 43], and constitutional classifiers are trained against universal jailbreaks [34]. We use a guard model only as an evaluation judge, since refusal-string matching mistakes harmless non-compliance for a successful attack [37].

Calibration under shift. Classical label-shift corrections reweight or re-estimate priors from unlabeled target data [1, 19, 33]; test-time adaptation updates normalisation statistics on the fly [39]; conformal methods restore coverage under covariate or label shift given weights or labels [28, 38]. Unsupervised recalibration adjusts a classifier's outputs per subpopulation without labels [46], and multicalibration asks for calibration on every identifiable subgroup [12]. For jailbreak detectors specifically, Piet et al. [27] adapt text classifiers to drift by self-training, Azov et al. [4] find that re-thresholding often suffices when target labels are avail-

able, and Kim and Lee [15] combine register erasure with distribution-free calibration on labelled benign data. Our method differs in using no labels from the shifted framing and in operating per framing cluster of an activation gate.

3 Setup

Prompts. Harmful behaviours come from JailbreakBench, the standard text behaviours of HarmBench, and AdvBench; benign prompts come from the matched benign split of JailbreakBench, the safe prompts of XSTest, and 400 prompts of OR-Bench-Hard, which are designed to look harmful. After deduplication the corpus has 1,550 prompts. We split it 50/20/30 into train, validation and test *by behaviour*, stratified by source, so no wrapped version of a test request is ever seen in training. The test split holds 241 harmful and 225 benign requests.

Framings. Each request is wrapped by eight framing families that keep its text verbatim. Four are *seen*: the plain request, a persona template (AIM, DAN, "grandma"), refusal suppression, and prefix injection. Four are *unseen* and are used only in the final test: a fiction or screenplay frame, an indirect injection disguised as a retrieved document or forwarded e-mail, base64 encoding, and leetspeak. We additionally optimise GCG suffixes [48] against the undefended model for 20 test behaviours (80 steps, 128 candidates per step, 20-token suffixes).

Model and gates. The target is Qwen2.5-1.5B-Instruct [29] without a system prompt. All gates read layer 14 of 28 and, when they fire, add the same steering vector $\alpha\,\hat{r}$ to the residual stream at every position, where $\hat{r}$ is the normalised difference between mean harmful and mean benign activations of plain training requests and α is 0.4 times the mean activation norm. Differences between gates therefore come from the gating decision alone. We compare (i) a CAST-style condition vector [16]; (ii) a logistic-regression probe and (iii) a two-layer MLP probe on the final-token activation; (iv) ARETE, a geometric gate of our own that maps activations onto separate intent and context hyperspheres, learns harm caps on the intent sphere, and steers in the orthogonal complement of a framing subspace estimated from wrapper displacements (Appendix A); and (v) probes reading other token positions, introduced in Section 4. An always-on steer bounds what steering can achieve. Every gate is trained on seen framings of the training split and thresholded so that its false-positive rate on benign validation prompts of seen framings is 1%. This shared, held-out-quantile threshold is the distribution-free calibration of Kim and Lee [15].

Protocol. Every design decision in this paper (token position, layer, clustering features, thresholds) is made by *leave-one-framing-out* (LOFO) on the validation split: each seen wrapper in turn is treated as unseen, the gate is trained and calibrated on the remaining seen framings, and recall is measured on the held-out one. The unseen families and the test split are used once, after the configuration is fixed.

Metrics. Detection is reported as true-positive rate (TPR) on harmful prompts and false-positive rate (FPR) on benign prompts at the calibrated threshold, plus AUROC. For generations (greedy, 96 new tokens) we use Qwen3Guard-Gen-4B in response-moderation mode [30], given the *unwrapped* request and the response: attack success rate (ASR) is the fraction of harmful requests whose response is judged UNSAFE, and over-refusal is the fraction of benign requests whose response is judged a refusal. Refusal-string matching over-counts attacks: on base64-wrapped requests the model often decodes the payload incorrectly and answers a different, harmless question.

4 Diagnosis: The Gate Knows, the Threshold Doesn't

Recall collapses on unseen framings. Table 1 shows that every gate detects 83–97% of harmful prompts on seen framings at the 1% budget, except CAST, whose single condition direction already fails on prefix injection. On unseen framings recall falls to between 0% and 17%. The benign FPR on these framings also falls to nearly zero, which is the first hint that the threshold, not the separation, has moved: the whole score distribution of a new framing shifts below the threshold. Our ARETE gate is no exception. Its fiction AUROC of 0.40 is worse than chance, so its learned intent geometry does not carry over to a narrative frame.

Ranking survives; the threshold does not. If a framing erased the harmful signal, no threshold could recover it. Table 2 tests this with an oracle that re-thresholds each probe on benign prompts of the unseen framing itself. For a probe reading the mean of the

	Seen framings				Unseen framings			
Gate	Direct	Persona	Ref. supp.	Prefix inj.	Fiction	Indirect	Base64	Leetspeak
CAST	85 (0.98)	42 (0.97)	50 (0.96)	0 (0.94)	0 (0.87)	0 (0.94)	0 (0.46)	0 (0.93)
MLP probe	91 (1.00)	90 (1.00)	87 (1.00)	83 (1.00)	0 (0.81)	0 (0.93)	0 (0.41)	0 (0.94)
ARETE (v1 geometry)	93 (0.99)	93 (0.99)	94 (0.99)	88 (0.99)	0 (0.40)	3 (0.71)	0 (0.47)	0 (0.93)
Probe, last token	94 (1.00)	91 (1.00)	90 (1.00)	87 (1.00)	0 (0.86)	15 (0.89)	0 (0.45)	0 (0.95)
Probe, user tokens (mean)	96 (1.00)	88 (1.00)	83 (1.00)	93 (1.00)	5 (0.99)	17 (0.98)	0 (0.53)	0 (0.72)
Multi-view union	97 (1.00)	93 (0.99)	92 (0.99)	88 (1.00)	0 (0.83)	12 (0.89)	1 (0.54)	5 (0.94)

Table 1: Detection on the test split at a 1% benign FPR budget calibrated on seen framings: TPR (%) on harmful prompts and AUROC in parentheses. All gates read layer 14 of Qwen2.5-1.5B-Instruct. Recall collapses on unseen framings even where AUROC stays high.

Readout	Framing	AUROC	TPR (shared)	TPR (oracle)
Last token	Fiction	0.86	0.0	21.6
	Indirect	0.89	15.4	41.9
	Base64	0.45	0.0	0.0
	Leetspeak	0.95	0.0	17.8
User mean	Fiction	0.99	4.6	80.5
	Indirect	0.98	16.6	78.8
	Base64	0.53	0.0	0.0
	Leetspeak	0.72	0.0	17.4

Table 2: Unseen framings, test split. *Shared*: threshold calibrated on seen framings (1% FPR). *Oracle*: threshold re-calibrated to 1% FPR on benign prompts of the same framing, which is unavailable in practice. Numbers are TPR (%).

user's tokens, AUROC on fiction and indirect injection is 0.99 and 0.98, and the oracle threshold lifts recall from 5% to 81% and from 17% to 79%. The information is present; the calibration is wrong. Base64 is the exception that proves the rule: AUROC is near chance and no threshold helps, because a 1.5B model does not reliably decode the payload, so its representation of the request is not formed in the first place.

Projecting out framing does not help. ARETE's premise is that framing occupies a low-dimensional subspace that can be removed. We test the premise directly in LOFO on the validation split, estimating the subspace from wrapper displacements of the remaining seen wrappers and projecting it out before training a linear probe. Removing 0, 4, 16 or 64 directions gives mean held-out recall of 54%, 54%, 54% and 48%. A subspace learned from two wrappers does not contain the shift induced by a third. Prefix injection is the clearest case, with recall near 1% at every rank.

Where to read. Token position matters more than projection. In LOFO, a probe on the last template token misses prefix injection (1%) because the injected instruction follows the request and changes the final state. A probe on the mean of the user's tokens catches 60% of it but is weaker on personas. A probe on the last user token fails on prefix injection for the same positional reason. This is consistent with harmfulness being represented over the user's tokens rather than only at the decision point [9, 45]. A union of the three readouts, each calibrated at a third of the budget, raises LOFO recall from 54% to 69% at 1.3% FPR, and on the test split it improves seen framings slightly (Table 1). It does not fix unseen framings, because each of its readouts suffers the same threshold shift.

5 Label-Free Framing-Adaptive Recalibration

The diagnosis suggests treating each framing as its own calibration problem. At deployment the framing of a new prompt is unknown and no labels are available, but two facts can be exploited. Prompts that share a framing share an activation signature that is easy to cluster. And real traffic is mostly benign, so within a cluster the bulk of the score distribution describes benign prompts.

Algorithm 1 formalises this. The gate score s is the user-token mean probe; the clustering features ϕ are mean-pooled user-token activations, reduced to 32 principal components; k ranges over $\{2, \ldots, 12\}$. Clusters with fewer than 30 prompts fall back to validation statistics. On the test traffic window silhouette chose k=12, the top of the range, so framings are split more finely than the eight families, which is harmless for a per-cluster threshold. The median and MAD tolerate contamination by a minority of harmful prompts, so a harmful prompt stands out against the benign bulk of its own framing rather than against the benign bulk

Algorithm 1 Framing-adaptive recalibration

Require: gate score $s(\cdot)$; framing features $\phi(\cdot)$; benign FPR budget ϵ

1: **Offline (validation):** simulate streams of seen framings; for each, run lines 2–4 and collect robust z of benign prompts; set $z^\star$ to their $(1-\epsilon)$ quantile.
2: **Online:** on a window of incoming prompts $\{x_i\}$, compute $\phi(x_i)$, reduce by PCA, run k-means with k chosen by silhouette.
3: For each cluster c: $m_c \leftarrow \mathrm{median}\{s(x_i)\}$, $d_c \leftarrow 1.4826 \cdot \mathrm{MAD}\{s(x_i)\}$.
4: Fire on x in cluster c iff $(s(x) - m_c)/d_c > z^\star$.

of the training framings. The threshold $z^\star$ is calibrated on validation streams that pass through the same clustering, so it accounts for within-cluster dispersion. We also evaluate a *hybrid* rule that fires if either the shared threshold or the adaptive threshold fires, each at half the budget. It is meant as a floor for regimes where the adaptive statistics are unreliable.

All choices were made by LOFO on validation streams at 5% harmful prevalence. The score readout was chosen from {last token, last user token, user mean} and the clustering layer from $\{7, 14, 21\}$. With the user-mean readout and clustering at layer 14, held-out recall rose from 38% (shared threshold) to 91% at 1.1–1.2% FPR. The other readouts did not benefit, since their failure on held-out wrappers is not only a matter of location. This configuration was then fixed and run once on the test split.

6 Results

Detection. Table 3 shows that at 1–5% prevalence adaptive recalibration lifts recall on unseen fiction prompts from 5–7% to 85–87% and on indirect injection from 17–20% to 65–70%, close to the oracle of Table 2. On these two framings the benign FPR stays at 0.6–1.0%. The gains are not free. The mean FPR over all framings rises from 0.5% to 1.6–1.8%, mostly from the base64 cluster, where scores are near noise and standardisation amplifies it (4.9% FPR), and from plain requests (about 3%). Recall on plain requests also drops from 95–98% to 80–88%. The hybrid rule recovers plain-request recall (93–98%) while keeping almost all of the unseen-framing gain. Leetspeak, on which even the oracle reaches only 17%, is not helped.

Generation. Detection gains matter only if they change what the model says. Table 4 reports judged attack success when each gate controls the same refusal steer. The always-on steer bounds what steering can do: it lowers fiction ASR from 88.8% to 28.6% but refuses every plain benign request. The existing gates sit at the other extreme. On fiction and indirect injection they never fire, so their ASR equals the undefended model's (88.8% and 12.4–12.9%). With per-cluster statistics estimated once from a traffic window and frozen, the adaptive gate fires on 86% of harmful fiction and 92% of harmful indirect prompts, and on 0.9% of benign prompts in those framings. This brings fiction ASR down to 34.4%, within six points of the always-on bound, and indirect ASR down to 4.6%. Over-refusal on plain benign requests rises by only 0.9 points, from 59.1% to 60.0%. The high base rate reflects how often the 1.5B model refuses the borderline prompts of XSTest and OR-Bench even without any defense. On seen framings the adaptive gate matches the best fixed-threshold gates, for example 12.0% versus 12.9–13.7% on prefix injection. Base64 is unaffected because the gate cannot see it (Section 4). Leetspeak never succeeds against this model in the first place.

Prevalence. The method assumes that harmful prompts are a minority of each framing cluster. Table 3 probes this assumption. At 10% prevalence adaptive recall on fiction is still 78%. At 20% the median and MAD are visibly contaminated and recall on every framing drops, to 28% on plain requests. At 35% (not shown) adaptive recall is at most 5% on every framing. The hybrid rule keeps plain-request recall at 95% even at 20% prevalence, because its shared half does not depend on traffic statistics. An attacker who can flood a framing cluster with harmful requests can therefore disable the adaptive half, but not the shared half.

7 Discussion and Limitations

What the results say. Activation gates are usually evaluated by AUROC, and by that measure they generalise well to new framings. Deployed gates are judged at a threshold, and there they do not. The gap between the two is a calibration problem, and it can be addressed without new labels by letting the threshold follow the framing. This is consistent with recent observations that probes keep their ranking but lose their operating point off-distribution [4, 15, 21, 23]. It also suggests that defenses which try to remove framing from the representation, as ARETE did, are solving the wrong problem when the shift is in location rather than in separability.

		Seen framings (TPR)				Unseen framings (TPR)				
π	Rule	Direct	Persona	Ref. supp.	Prefix inj.	Fiction	Indirect	Base64	Leetspeak	FPR
1%	Shared	98	92	80	92	5	20	0	0	0.5
	Adaptive	88	95	100	92	85	70	0	5	1.8
	Hybrid	98	95	100	92	85	68	0	5	1.7
5%	Shared	95	86	85	91	7	17	0	0	0.5
	Adaptive	80	90	94	88	87	65	0	4	1.6
	Hybrid	93	88	94	87	86	64	0	2	1.5
10%	Shared	96	89	83	92	4	17	0	0	0.5
	Adaptive	73	87	86	82	78	56	0	1	1.3
	Hybrid	94	86	84	82	76	54	0	1	1.3
20%	Shared	97	87	83	92	5	17	0	0	0.5
	Adaptive	28	50	61	56	52	36	0	1	0.8
	Hybrid	95	59	76	81	49	35	0	0	1.0

Table 3: Detection on the test split for unlabeled streams with harmful prevalence π (20 random draws each; the stream contains all benign test prompts of every framing and harmful prompts subsampled to prevalence π). TPR (%) per framing and benign FPR (%) averaged over all eight framings. Same probe throughout; only the decision rule differs.

	Seen framings				Unseen framings					
Defense	Direct	Persona	Ref. supp.	Prefix inj.	Fiction	Indirect	Base64	Leetspeak	GCG	Over-ref.
None	1.2	4.1	2.9	75.9	88.8	12.9	51.0	0.0	25.0	59.1
Always-on steer	0.4	0.4	0.4	4.6	28.6	0.4	0.0	0.0	0.0	100.0
CAST	0.8	4.6	2.9	75.9	88.8	12.9	51.0	0.0	25.0	59.6
Linear probe (last)	0.4	2.1	1.7	13.7	88.8	12.4	51.0	0.0	10.0	59.1
ARETE	1.9	1.7	1.7	33.3	88.8	12.9	51.0	0.0	21.7	59.1
Multi-view union	0.4	1.2	1.2	12.9	88.8	12.9	50.6	0.0	15.0	59.6
Adaptive recal. (ours)	1.2	1.2	1.2	12.0	34.4	4.6	50.6	0.0	–	60.0

Table 4: Generation-level results on the test split, judged by Qwen3Guard-Gen-4B. Columns 2–10: attack success rate (%, response judged UNSAFE); GCG uses 20 behaviours. Last column: over-refusal (%) on plain benign requests. Gated rows average three seeds. For the adaptive rule, per-cluster statistics are estimated once from an unlabeled traffic window (all benign test prompts plus harmful prompts at 5% prevalence) and then frozen and applied to every test prompt through its nearest cluster centroid. GCG prompts were not part of that window.

Limitations. Our experiments use a single 1.5B-parameter target model. Planned experiments on a 7B model could not be run because GPU capacity was unavailable. Larger models parse encodings such as base64, which would likely turn that framing from a capability failure into a calibration failure. The framing families are template-level; adaptive optimisation against the gate itself [5] is out of scope, and an adaptive attacker can also target the clustering or flood a cluster, as the prevalence study shows. The judge is a model, and on wrapped benign prompts it often labels confused non-compliance as refusal, so we report over-refusal mainly for plain benign requests. Finally, our GCG evaluation uses 20 behaviours and is indicative only.

8 Conclusion

Activation-gated jailbreak defenses fail on unseen framings not because the model stops representing harm, but because the gate's threshold stays where the developers put it. Recalibrating that threshold per framing cluster, using only the robust statistics of unlabeled traffic, recovers most of the lost recall at near-constant false-positive cost, with failure modes that we have mapped and that a simple hybrid rule partly contains.

Ethics Statement

This work studies defenses against misuse of language models. All harmful prompts come from public bench-

marks; we release no new harmful content, and generated responses are stored only for evaluation. The attack templates we use are publicly documented. Our negative results, including the prevalence-flooding failure mode, are reported so that defenders do not overtrust activation gates.

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A ARETE Geometry

ARETE standardises activations, maps them through two learned linear heads onto unit hyperspheres for intent and context, and trains with (i) an invariance loss pulling the intent coordinates of a wrapped prompt towards those of its plain version, (ii) a framing-classification loss on the context sphere, and (iii) a cap loss that places harmful intents inside, and benign intents outside, a set of eight unsupervised spherical caps. The gate score is the maximum cap margin. The steering direction is the refusal direction projected onto the orthogonal complement of a rank-16 framing subspace estimated by SVD of wrapper displacements $h_{\text{wrapped}} - h_{\text{plain}}$ on training behaviours.